Akoizo Original Practice - Form 2 - Answer key

Division C | Experimental Design | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

ANSWER KEY - Review only after submitting.

1. (2 points) Which is a controlled variable rather than a control treatment?

Unit: Problem, hypothesis, variables, and experimentation.
Worked-case data: All low-level trials occur early and all high-level trials occur late while the instrument drifts upward.

A. A baseline reference level

B. An untreated comparison

C. Keeping light constant

D. A no-added-fertilizer group

Correct: C

2. (2 points) A standard of comparison is…

Unit: Division C comparison standards, significant figures, and abstracts.
Worked-case data: Report area from measured lengths 2.0 cm and 3.00 cm, then report their sum.

A. A defined reference condition

B. Every controlled variable

C. Always a zero response

D. Always the largest treatment

Correct: A

3. (2 points) Standard deviation has…

Unit: Quantitative data, graphs, and statistics.
Worked-case data: For response values 8, 10, 12, calculate mean, range, and sample SD.

A. The independent-variable units necessarily

B. The response’s units

C. Squared response units

D. No units always

Correct: B

4. (2 points) A failed hypothesis means the experiment is invalid?

Unit: The experiment and the prompt packet.
Worked-case data: A prompt supplies a ramp, object, ruler, and timer. Turn “study motion” into a testable question.

A. Only if data vary

B. Only if the graph is flat

C. No

D. Yes

Correct: C

5. (2 points) A conclusion should answer…

Unit: Errors, conclusions, and recommendations.
Worked-case data: Every timer reading is 0.5 s too high. Would ten more repeats eliminate the bias?

A. The original measured question

B. Every possible mechanism

C. A preferred hypothesis regardless of data

D. A different untested outcome

Correct: A

6. (4 points) Improve “release and time it.”

Unit: Materials, reproducible procedures, and qualitative data.
Worked-case data: A trial has no timer reading because the observer missed the endpoint. What should the table and procedure record?

answer: Specify release location and method, start/stop marks, timer operation, reset, trial count, and recording units. (4 points).

7. (4 points) Name three dependent reporting tasks.

Unit: Practice, timing, teamwork, and final review.
Worked-case data: A team finishes data collection with little time left and no graph or analysis. Suggest a targeted practice change.

answer: Statistics, graphing, and conclusions depend on collected measurements; the abstract depends on completed analysis. (4 points).

8. (4 points) Give a variance mechanism for timing.

Unit: Claims, evidence, and reasoning for trends and outliers.
Worked-case data: At levels 1,2,3, means are 12,16,20 units, with SD 2 at each level. Write a bounded trend CER.

answer: Inconsistent start/stop reaction can vary measured duration across trials, increasing spread. (4 points).

9. (4 points) Why use several levels?

Unit: Problem, hypothesis, variables, and experimentation.
Worked-case data: All low-level trials occur early and all high-level trials occur late while the instrument drifts upward.

answer: They reveal the response pattern over a range rather than only a single pairwise difference. (4 points).

10. (4 points) Write a compact abstract skeleton.

Unit: Division C comparison standards, significant figures, and abstracts.
Worked-case data: Report area from measured lengths 2.0 cm and 3.00 cm, then report their sum.

answer: Question and system; levels, response, and replication; key quantitative result with variability; conclusion and main limitation. (4 points).

11. (4 points) Why can precise measurements be inaccurate?

Unit: Quantitative data, graphs, and statistics.
Worked-case data: For response values 8, 10, 12, calculate mean, range, and sample SD.

answer: A systematic offset can shift all tightly clustered readings away from the reference. (4 points).

12. (6 points) Solve this case and explain your reasoning: A prompt supplies a ramp, object, ruler, and timer. Turn “study motion” into a testable question.

reasoning: Choose release height as the independent variable with defined levels.
Define response as time between two fixed marks, with units and timing endpoints.
Hold object, ramp surface, release method, and travel distance consistent and replicate each level. (3 points).

conclusion: A narrow measurable comparison supports a reproducible experiment. (3 points).

13. (6 points) Solve this case and explain your reasoning: Every timer reading is 0.5 s too high. Would ten more repeats eliminate the bias?

reasoning: The offset is systematic.
Averaging more readings can reduce some random variation but retains the 0.5 s offset.
Calibrate or correct using a justified reference method and document the change. (3 points).

conclusion: Replication alone cannot remove a known systematic offset. (3 points).

14. (2 points) Define “Qualitative data” in the context of Experimental Design.

answer: Descriptive observations rather than the primary numerical measurement. (2 points).

15. (2 points) Define “Bottleneck” in the context of Experimental Design.

answer: Task limiting overall progress or quality. (2 points).

16. (2 points) Define “Evidence” in the context of Experimental Design.

answer: Relevant observations and quantitative results. (2 points).

17. (2 points) Define “Replication” in the context of Experimental Design.

answer: Repeated independent experimental units or trials as specified. (2 points).

18. (2 points) Define “Significant figures” in the context of Experimental Design.

answer: Digits communicating measurement precision under a convention. (2 points).